

Chapter 3

Neutron Scattering and Highly Frustrated Magnetism

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Abstract Neutron scattering is a paradigm technique for the determination of spin correlation functions. It is the method of choice for the study of magnetic order and magnetic interactions, and, in favourable cases, it may provide an unambiguous distinction between conventional paramagnetism and cooperative paramagnetism. This chapter is an introduction to neutron scattering with an emphasis on its interpretation in terms of spin correlation functions and generalised susceptibilities. The main aim is to illustrate how neutron scattering is used to interrogate the nature of the disordered and highly degenerate states that occur in highly frustrated magnets.

3.1 Introduction

The study of the phases of matter might be seen, at a fundamental level, as the study of broken symmetries, of which there are many kinds [1]. Magnetic neutron scattering essentially measures the spin–spin correlation function, so most clearly distinguishes magnetic phases that have correlation functions of differing symmetry. The simplest example is the distinction of a paramagnet and a spin ordered phase (ferromagnet or antiferromagnet). In the former the spin–spin correlation is a decreasing function of distance between the spins, which gives rise to continuous or ‘diffuse’ neutron scattering in reciprocal space; in the latter the correlation function acquires a component with periodic symmetry that gives rise to discrete scattering in the form of Bragg peaks. Thus, given sufficient instrumental resolution, there is no ambiguity in distinguishing ordered and disordered spin components as their signatures in neutron scattering are radically different (Fig. 3.1).

The game of highly frustrated magnetism is to find new and exotic phases, that may generally be described as ‘cooperative paramagnets’ [2]. Bulk measurements may infer the existence of such phases, but they do not generally indicate a distinctive difference between these and the conventional paramagnetic phase, so we should ask, is it even correct to imagine a basic distinction between the two? If the answer were no, then highly frustrated magnetism would be a very boring subject! Fortunately, there is such a basic distinction, at least for one broad class of highly

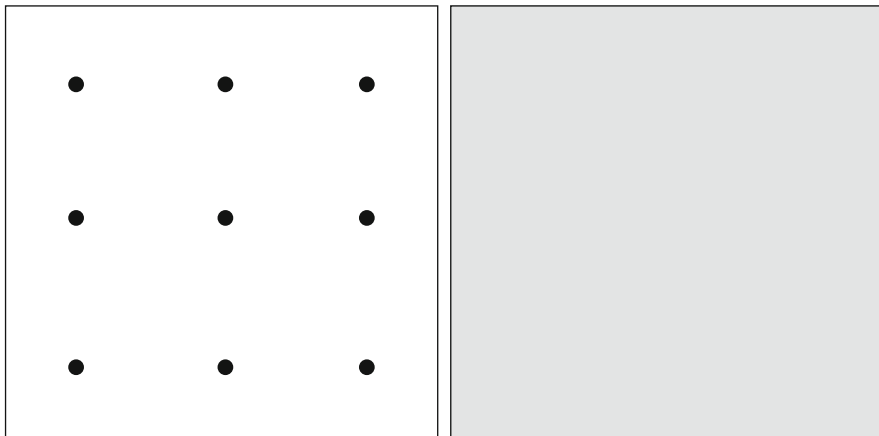


Fig. 3.1 Schematic neutron scattering map of an ordered phase with sharp Bragg spots (*left*) and an ideal paramagnetic phase, with broad diffuse scattering (*right*)

frustrated magnets, and this is clearly revealed by neutron scattering. The clarity of the distinction afforded by neutron scattering again reflects symmetry differences in the underlying spin correlation functions of a conventional and a cooperative paramagnet, even though neither phase exhibits periodic spin order.

To emphasise this, Fig. 3.2 illustrates reciprocal space maps of the correlation function $S(\mathbf{Q})$ (defined below), a quantity directly estimable by neutron scattering. Assuming a cubic system, the maps may be parameterised in polar coordinates $\{q, \theta\}$ with respect to a Brillouin zone centre. We consider four possibilities: (a) that S depends neither on q nor θ , (b) that it depends on q but not θ , (c) that it depends on θ but not q , and (d) that it depends on both. In fact, this classification turns out to correspond respectively to the case of (a) an ideal paramagnet, (b) a conventional magnet in its paramagnetic phase, (c) an ideal highly frustrated magnet with ‘emergent gauge’ structure, and (d) a more general highly frustrated magnet. This correspondence is a vivid demonstration of how the most straightforward classification of neutron maps naturally identifies distinct paramagnetic phases that fit neatly with our concepts of frustration.

In this article, I want to elaborate upon these remarks in order to explain how neutron scattering may be used (and has been used so far) to interrogate the nature of cooperative paramagnetic phases. This interrogation may be direct – by measuring spin–spin correlations – or indirect: by establishing the spin–spin interactions, typically through characterisation of magnetic structures, spin waves or crystal field excitations. My additional aim is to give the reader a basic introduction to neutron scattering and its relation to statistical mechanics.

The article is divided into four sections. The first Sect. 3.2 describes what neutron scattering measures (within the kinematic theory, stemming from the first Born approximation) and how it relates to spin correlation functions, generalised susceptibilities and magnetic moments. The second Sect. 3.3 contrasts the signatures,

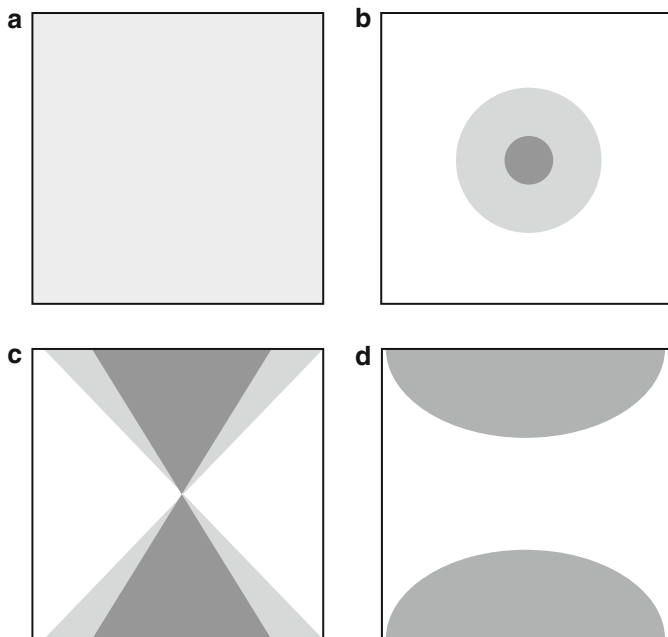


Fig. 3.2 Schematic neutron scattering maps classified by their radial and angular dependence near to a Brillouin zone centre: see text for the definition of (a)–(d)

in neutron scattering, of conventional and highly frustrated magnetism. The third Sect. 3.4 surveys some experimental results: these are chosen to illustrate general principles, rather than to exhaustively review the subject. The final Sect. 3.5 draws some conclusions. I do not discuss the generation of neutron beams or any aspects of neutron instrumentation and I make no attempt to describe the full array of neutron scattering techniques.

3.2 What Neutron Scattering Measures

3.2.1 Scattering Triangle

The basic neutron scattering experiment is illustrated in Fig. 3.3. An incident neutron of known wavevector $\mathbf{k}_i$ is scattered by the sample into an outgoing (final) neutron of wavevector $\mathbf{k}_f$.¹ It is possible to measure the change in both the direction and magnitude of the neutron wavevector, or equivalently, the changes in the

¹ More precisely, one should conceive of a probability current of incoming and scattered neutrons.

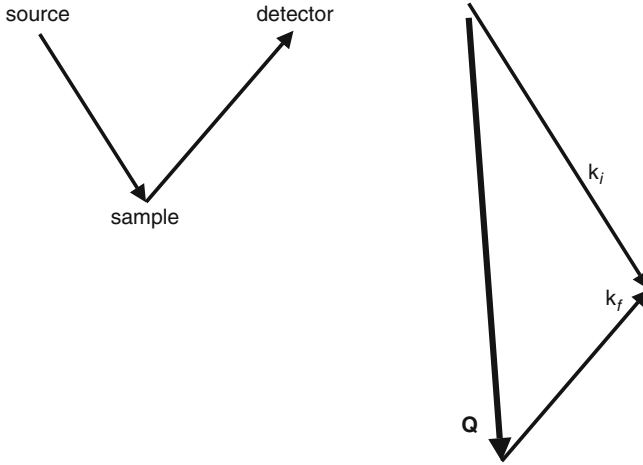


Fig. 3.3 Schematic neutron scattering experiment (*left*) and one possible scattering triangle that is consistent with the scattering angle shown

neutron's momentum $\Delta \mathbf{p} = \hbar \Delta \mathbf{k}$. The conservation of momentum leads to the vector relationship:

$$\Delta \mathbf{p} / \hbar = \mathbf{k}_i - \mathbf{k}_f = \mathbf{Q}, \quad (3.1)$$

which is summarised in the scattering triangle, Fig. 3.3. The wavevector $\mathbf{Q}$ is called the *scattering vector*. If $|\mathbf{k}_i| \neq |\mathbf{k}_f|$ then the triangle is non-isosceles and there is a change in neutron energy $\Delta E = \hbar^2 \Delta(k^2) / 2m$, where m is the neutron mass. This may be expressed as a frequency ω :

$$\Delta E = \frac{\hbar^2}{2m} (k_i^2 - k_f^2) = \hbar \omega, \quad (3.2)$$

where $\Delta E = \hbar \omega$ is called the *energy transfer*. The scattering vector and energy transfer are the two key parameters in neutron scattering that can be selected by the experimenter.

The usual neutron experiment delivers a beam of unpolarised neutrons and detection does not discriminate between scattered spin states. However, neutrons may also be delivered in a polarised state with analysis of the scattered neutron spin states. In this way, changes in spin angular momentum may also be characterised.

3.2.2 Partial Differential Cross Section

Like any triangle, the scattering triangle can be fully specified in terms of three variables, which we can choose in different ways. For example, we might specify the incoming wavelength, scattering angle and change in energy. In order to detect

neutrons we need to collect the scattered neutrons over a range of angles in three dimensions, i.e., a range of solid angles $\Delta\Omega$, as well as a range of scattered neutron energies ΔE . We can conceive of an ideal high resolution experiment in which these ranges are small and can be taken as differential quantities, $d\Omega$ and dE . The number of neutrons scattered per second per unit incident flux, into a range of solid angle $d\Omega$ and with range of energies between E and $E + dE$, is called the partial differential scattering cross section, $d^2\sigma/d\Omega dE$. We can expect this quantity to be the product of several terms. First, a characteristic length squared to give it the right dimensions – call it r_0^2 ; second, a term (k_f/k_i) because it depends on scattered/incident number of neutrons per unit time, quantities which are proportional to the velocity of the neutrons and hence to k ; and third, some function (say ϕ) of $\mathbf{Q}$ and ω as well as the possible spin states of the neutrons:

$$\frac{d^2\sigma}{d\Omega dE} = r_0^2 \frac{k_f}{k_i} \phi(\mathbf{Q}, \omega, \text{spin}). \quad (3.3)$$

3.2.3 Relation to Sample Properties

Neutrons are scattered by atomic nuclei (‘nuclear scattering’) and magnetic moments or spins (‘magnetic scattering’). The key feature of neutron scattering is that because momentum and energy are conserved quantities, then the changes in neutron linear momentum ($\hbar\mathbf{Q}$), neutron angular momentum (s) and neutron energy ($\hbar\omega$) are exactly mirrored by momentum and energy changes in the sample. Two examples from nuclear scattering illustrate this. The first is phonon scattering where the neutron excites or de-excites a phonon of wavevector $\mathbf{q} = \mathbf{Q}$ and frequency $\omega_0 = \omega$. Thus, scattering is only observed at those values of $\{\mathbf{Q}, \omega\}$ that coincide with the phonon ‘dispersion’ curve $\omega_0(\mathbf{q})$. The second example is Bragg reflection. Here, the scattering is *elastic* ($\omega = 0$ so $|\mathbf{k}_i| = |\mathbf{k}_f|$), and *coherent*,² so produces diffraction: because the angle of incidence is equal to the angle of ‘reflection’, the momentum transfer $\hbar\mathbf{Q}$ is normal to the set of crystal planes (hkl) doing the diffracting. By applying Bragg’s law it is easily shown that diffraction may only occur when $\mathbf{Q} = \mathbf{G}$ where $\mathbf{G}$ is a reciprocal lattice vector with $|\mathbf{G}| = 2\pi/d_{hkl}$ (here d is the interplanar spacing).

One can see from these examples that any physical property of the sample that can be represented in terms of wavevectors and frequencies may be directly observable to neutron scattering. Thus, neutron scattering directly looks at reciprocal space (the set of all wavevectors) and frequency space (the set of all frequencies) and most clearly reveals sample properties like phonons and crystal planes that are naturally represented in these spaces.

² Bragg scattering is coherent in the absolute sense, which means that the scattered beam interferes with the incoming beam. Diffuse scattering, treated below, may only be coherent in the relative sense which means that only the scattered waves produce interference effects [3].

3.2.4 Scattering from Atomic Magnetic Moments

Here, we are interested in magnetic neutron scattering from the electrons in a sample and I will assume that the nuclear scattering and other unwanted scattering (from say nuclear spins or the sample environment) can always be removed to reveal only this component (in reality this may be tricky in a powder experiment but it is usually straightforward in a single crystal experiment). Also, when I speak of ‘spins’ it is to be understood that these could be effective spin operators that describe the combined spin and orbital angular momentum.³

The neutrons are scattered by the spatially varying magnetic field $\mathbf{B}(\mathbf{r})$ that arises from the spins in the sample. Neutron scattering sees the Fourier transformed field $\mathbf{B}(\mathbf{q})$ which is proportional (within the constraint discussed below) to the Fourier transformed spin field $\mathbf{S}(\mathbf{q})$. Interference effects are produced by the relative coherence in the scattering from different atoms and bearing in mind that the spin field can vary in time as well as space, second order dynamical perturbation theory shows that the scattering is essentially proportional to the time (t) and spatial Fourier transform of the two-spin correlation function of the sample:⁴

$$S^{\alpha\beta}(\mathbf{q}, \omega) = \frac{1}{2\pi} \sum_{\mathbf{R}} \int_{-\infty}^{+\infty} e^{i\mathbf{q}\cdot\mathbf{R}-\omega t} \langle S_0^\alpha(0) S_{\mathbf{R}}^\beta(t) \rangle dt, \quad (3.4)$$

where $\mathbf{R}$ defines the spin positions, $\langle \dots \rangle$ denotes a thermal average and $\alpha, \beta = x, y, z$ are cartesian components. $S^{\alpha\beta}(\mathbf{q}, \omega)$ (called, in the context of neutron scattering the ‘dynamic scattering function’) is the sample property that the neutron scattering experiment seeks to estimate. However, as explained below, only some of the components of this tensor can be observed at any particular setting of the neutron instrument.

3.2.5 Orientation Factor and Form Factor

The observed scattering is modified by several extra factors, of which two are particularly important. The first of these is a direct consequence of Maxwell’s Law $\nabla \cdot \mathbf{B} = 0$ that implies that $\mathbf{B}(\mathbf{q})$ has no component parallel to $\mathbf{q}$.⁵ This means that neutrons only ‘see’ spin components perpendicular to the scattering vector $\mathbf{Q} = \mathbf{q}$. Consequently, the scattering function $S^{\alpha\beta}(\mathbf{Q}, \omega)$ must be multiplied by

³ Atomic magnetic moments may always be treated as proportional to effective angular momentum operators: the so-called *magnetomechanical equivalence*.

⁴ Small $\mathbf{q}$ will be used to represent sample properties: it is always equal to the vector $\mathbf{Q}$ of the scattering experiment, but there is a logical distinction.

⁵ The Fourier transform of $\nabla \cdot \mathbf{X} = 0$ is $\mathbf{q} \cdot \mathbf{X}(\mathbf{q}) = 0$, a result that we shall use again below (here $\mathbf{X}$ is a vector field).

an orientation factor that projects out its parallel components. This factor may be written $\delta_{\alpha\beta} - \hat{Q}_\alpha \hat{Q}_\beta$ where $\hat{Q}_{\alpha,\beta}$ are components of the unit vector $\hat{\mathbf{Q}} = \mathbf{Q}/|\mathbf{Q}|$.

The second modification arises from the fact that the unpaired spins actually inhabit spatially extended valence orbitals, so neutrons are really scattered by spin density rather than spins localised on lattice points. Fortunately, there is a rigorous theorem – the Wigner–Eckart theorem – that shows that to within a factor the spin density on a given atom or ion may be replaced with an effective spin operator localised on a lattice point [4]. $S^{\alpha\beta}(\mathbf{Q}, \omega)$ is derived from these localised operators so must be corrected to take into account the spatially extended spin density. The correction factor is $f(\mathbf{Q})^2$ where f is the magnetic form factor, which is essentially the Fourier transform of the normalised spin density on a single atom or ion (and hence depends on atom or ion *type*). Because ions are approximately spherical, it is typically a simple decreasing function of $|\mathbf{Q}|$ which (at least for our purposes) may be treated as a experimental correction to be looked up in tables [5].

3.2.6 General Expression for the Neutron Scattering

Collecting together the factors discussed above and recalling (3.3), the scattered intensity may be written

$$\frac{d^2\sigma}{d\Omega dE} = r_0^2 \frac{k_f}{k_i} f(\mathbf{Q})^2 \sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{q}_\alpha \hat{q}_\beta) S^{\alpha\beta}(\mathbf{Q}, \omega) \quad (3.5)$$

where the summation over α, β is specific to an unpolarised neutron experiment (see below). The factor r_0 , called the magnetic scattering length, depends on a number of fundamental constants that define the interaction of the neutron with the atomic magnetic moment: specifically, $r_0 = \gamma_n e^2 / m_e c^2 = -0.5391 \times 10^{-14}$ m, where γ_n is the neutron gyromagnetic ratio [6]. Equation (3.5) should also be corrected by a Debye–Waller type thermal factor that describes the blurring of the lattice due to thermal vibrations. The equation as written may be applied directly to the scattering from a single crystal sample: with a polycrystalline sample one further needs to average over crystallite orientations, which obscures much useful information.

If necessary, the components of $S^{\alpha\beta}(\mathbf{Q}, \omega)$ can be separated by the technique of polarisation analysis [7], thus removing the summation in (3.5). As the neutron has spin $s = 1/2$, the magnetic scattering might either flip this spin or not, with spin angular momentum conserved by a corresponding change in the spin state of the sample. Polarisation analysis measures spin flip and non spin flip cross sections as a function of incident neutron polarisation, from which the components of $S^{\alpha\beta}$ can be reconstructed [8].

3.2.7 *Real Experiments*

In a real experiment, in addition to applying (3.5), one needs to account for instrumental resolution [9, 10], sample imperfection, absorption by the sample and counting statistics. It is usual to assume a parameterised form of $S^{\alpha\beta}(\mathbf{Q}, \omega)$, then to convolve this with a parameterised resolution function (that includes the ‘mosaic spread’ of the sample), and finally to least-squares fit to the (possibly absorption-corrected) neutron scattering data. The experimental statistics in neutron scattering are strictly Poissonian, i.e., an error of $\sqrt{n}$ on count n , which renders this sort of approach to data analysis particularly robust. It should also be noted that (3.5) is only valid for a sufficiently small or ‘ideally imperfect’ sample, which naturally avoids the problem of multiple scattering and consequent extinction effects. The typical mosaic structure of real crystals is an advantage here as the coherent scattering volume is limited to a few microns. Generally speaking, the maximum crystal size should be a few mm^3 for Bragg scattering and a few cm^3 for diffuse scattering. Within this constraint the crystal volume should be maximised, to optimize counting statistics.

3.2.8 *Powder Averaging*

For most types of neutron experiment single crystal samples are preferable to powders. When a powder sample is used, there is averaging over the orientations of $\mathbf{Q}$ which obscures a great deal of information. However, there are certain situations in which powder samples may be preferred even if single crystals are available. For example, in magnetic crystallography, powder samples are often used, in order to cover more reciprocal space and to avoid extinction effects, which can affect Bragg intensities. Nevertheless, probably the most common reason for measuring powder samples is simply to maximise sample volume in cases where sufficiently large single crystals are not available, or neutron beam time is limited.

3.2.9 *Static Approximation*

In the case that energy transfers to and from the sample are negligible, the dynamic scattering function approaches a delta function in ω at the origin. Integration over frequencies then gives the so-called ‘static approximation’:

$$S^{\alpha\beta}(\mathbf{Q}) = \sum_{\mathbf{R}} \langle S_0^\alpha(0) S_{\mathbf{R}}^\beta(0) \rangle e^{i\mathbf{Q} \cdot \mathbf{R}} = \int_{-\infty}^{+\infty} S^{\alpha\beta}(\mathbf{Q}, \omega) d\omega. \quad (3.6)$$

The quantity on the left hand side of (3.6) is an instantaneous correlation function (i.e., one realised at time $t = 0$), but in a real experiment, such instantaneous correlations are sampled at a sequence of times that are widely separated on the time scale of spin fluctuations. This process is akin to taking an ergodic average, which should be well represented by an ensemble average in equilibrium statistical mechanics. The assumed fact that energy transfers are negligible means that the equilibrium state of the sample is not perturbed by the neutrons, so $S(\mathbf{q})$ measures a genuine equilibrium correlation function.

In this case, the differential cross section, which no longer depends on energy, becomes:

$$\frac{d\sigma}{d\Omega} = r_0^2 \frac{k_f}{k_i} f(\mathbf{Q})^2 \sum_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{Q}_\alpha \hat{Q}_\beta \right) S^{\alpha\beta}(\mathbf{Q}) \quad (3.7)$$

One may thus regard the total scattering at $\mathbf{Q}$ as an estimate of $S(\mathbf{Q})$.

3.2.10 Wavevector Dependent Magnetic Moment and Susceptibility

A very useful alternative formulation of (3.7) exploits the general relationship between correlation function, susceptibility χ and average magnetic moment M . For a sample with N spins located on a Bravais lattice of position vectors $\mathbf{R}$, we define the (intensive) wavevector dependent magnetic moment,

$$\mathbf{M}(\mathbf{q}) = N^{-1} \sum_{\mathbf{R}} S_{\mathbf{R}} e^{i\mathbf{q}\cdot\mathbf{R}} \quad (3.8)$$

from which follows

$$S^{\alpha\beta}(\mathbf{q}) = N \left\langle M^\alpha(\mathbf{q}) M^\beta(-\mathbf{q}) \right\rangle, \quad (3.9)$$

which may be verified from (3.8), by invoking the translational invariance of the sample (so $\sum_{\mathbf{R}_0} \sum_{\mathbf{R}} \rightarrow N \sum_{\mathbf{R}-\mathbf{R}_0}$). Defining the (intensive) wavevector dependent susceptibility:

$$\chi^{\alpha\beta}(\mathbf{q}) = \frac{dM^\alpha}{dH^\beta} = \frac{N}{kT} \left[\langle M^\alpha(\mathbf{q}) M^\beta(-\mathbf{q}) \rangle - \langle M^\alpha(\mathbf{q}) \rangle \langle M^\beta(-\mathbf{q}) \rangle \right], \quad (3.10)$$

we see that

$$S^{\alpha\beta}(\mathbf{q}) = kT \chi^{\alpha\beta}(\mathbf{q}) + N \langle M^\alpha(\mathbf{q}) \rangle \langle M^\beta(-\mathbf{q}) \rangle. \quad (3.11)$$

Neutron scattering thus measures both wavevector dependent magnetic moments and susceptibilities.⁶

⁶ This susceptibility is really a reduced susceptibility that should be multiplied the form factor squared to convert it to the ‘real’ susceptibility.

3.2.11 Fully Ordered Magnet

In a fully ordered magnet, the susceptibility components are all zero⁷ and all scattering is due to the wavevector dependent magnetic moment. Assuming that the crystal structure has the symmetry of a Bravais lattice (as is nearly always the case for geometrically frustrated magnets), then the spin positions $\mathbf{R}$ may be represented as $\mathbf{R} = \mathbf{T} + \mathbf{r}$ where $\mathbf{T}$ is a translation vector that defines a particular magnetic unit cell and $\mathbf{r}$ is the position of a magnetic atom or spin within a unit cell. If the wavevectors of the *magnetic* reciprocal lattice are $\mathbf{G}$ then $\langle \mathbf{M}(\mathbf{q}) \rangle$ can only be finite for $\mathbf{q} = \mathbf{G}$. At these wavevectors $\langle \mathbf{M}(\mathbf{q}) \rangle$ becomes $N^{-1} \sum_{\mathbf{T}} \sum_{\mathbf{r}} \langle \mathbf{S}(\mathbf{T} + \mathbf{r}) \rangle e^{i\mathbf{G} \cdot (\mathbf{T} + \mathbf{r})} = \sum_{\mathbf{r}} \langle \mathbf{S}(\mathbf{r}) \rangle e^{i\mathbf{G} \cdot \mathbf{r}}$, since $e^{i\mathbf{G} \cdot \mathbf{T}} = 1$ [3]. Plugging this result into (3.11), we find:

$$\frac{d\sigma}{d\Omega} = r_0^2 \frac{k_f}{k_i} f(\mathbf{Q})^2 \left| \sum_{\mathbf{r}} \langle \mathbf{S}_{\perp}(\mathbf{r}) \rangle e^{i\mathbf{Q} \cdot \mathbf{r}} \right|^2 \delta(\mathbf{Q} - \mathbf{G}), \quad (3.12)$$

where the orientation factor (which projects out parallel spin components) has been absorbed into the ‘ $\perp$ ’ symbol, $\mathbf{S}_{\perp}$ being the component, perpendicular to $\mathbf{Q}$, of a particular spin in the unit cell of the magnetic structure. The scattering is seen to occur as delta functions, or Bragg peaks, at the magnetic reciprocal lattice vectors, with intensity proportional to the square of the ordered component. In the above equation, the summation is over the unit cell rather than the whole sample and the term $|\dots|$ is referred to as the *magnetic structure factor* [11].

Magnetic Bragg peaks do not necessarily occur at all the magnetic reciprocal lattice points. There are two very general reasons for this. First, the orientation factor (the ‘ $\perp$ ’ in (3.12) might cause certain Bragg reflections to have zero intensity. Second, the phase factors $e^{i\mathbf{Q} \cdot \mathbf{r}}$ might cause the scattering from different spins in the unit cell to interfere destructively, leading to systematic absences. This always occurs if the magnetic and nuclear unit cells are non-coincident, causing the magnetic Bragg peaks to occur at different positions in reciprocal space to the nuclear Bragg peaks, as illustrated in Fig. 3.4. In the example, the magnetic unit cell is doubled with respect to the nuclear unit cell along one direction. The nuclear reciprocal lattice $\{\mathbf{G}_{\text{nuclear}}\}$ is a subset of the magnetic reciprocal lattice $\{\mathbf{G}_{\text{magnetic}}\}$, but the set $\{\mathbf{G}_{\text{nuclear}}\}$ are systematic absences for magnetic diffraction, leading to a neat separation of nuclear and magnetic Bragg peaks.

If we define $\{\mathbf{G}_{\text{magnetic Bragg}}\}$ as the set of magnetic Bragg peak positions then we can write

$$\mathbf{G}_{\text{magnetic Bragg}} = \mathbf{G}_{\text{nuclear}} + \sum_i \mathbf{k}_i \quad (3.13)$$

where the $\mathbf{k}_i$ s are a finite set of propagation vectors of the magnetic structure [11]. In the example (Fig. 3.4), the set $\{\mathbf{k}_i\}$ comprises only the single vector $\mathbf{k} = (0, 0, \frac{1}{2})$, but more generally, so-called multi- k structures are also possible [11]. As described

⁷ At least in the hypothetical case where the magnet has no possible fluctuations.

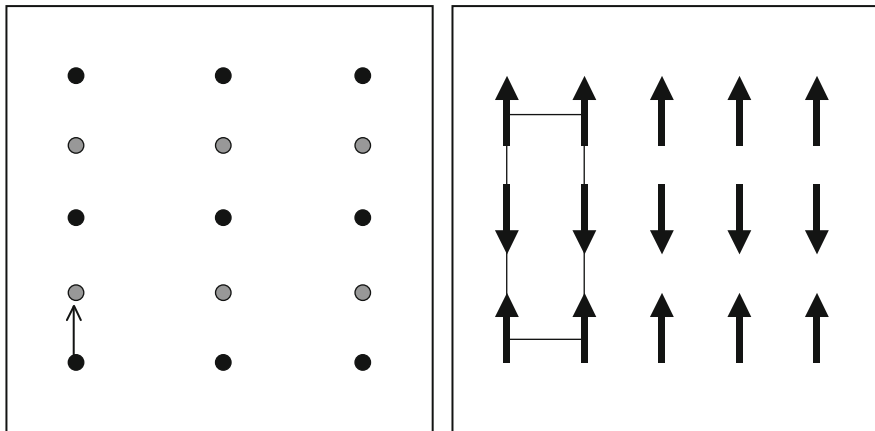


Fig. 3.4 The magnetic structure on the right (box is the magnetic unit cell which is twice the nuclear unit cell) produces to a diffraction pattern in reciprocal space like that shown on the left. Magnetic Bragg peaks (*grey circles*) appear intercalated between nuclear Bragg peaks (*black circles*) and are related to them by a propagation vector $\mathbf{k}$ (*arrow*)

later (Sect. 3.4), examples of a single- k and a four- k structure are provided by the rare earth pyrochlores $\text{Gd}_2\text{Sn}_2\text{O}_7$ [12] and $\text{Gd}_2\text{Ti}_2\text{O}_7$ [13] respectively.

In practice, magnetic structures are solved by fitting experimental data, via (3.12) to candidate models: group theoretic techniques (representational analysis) may be used to identify the possibilities.

3.2.12 Magnet with Full or Partial Disorder

Magnetic disorder is characterised by finite susceptibilities. In general, the wavevector dependent susceptibility is a smoothly varying function of wavevector so the scattering arising from the magnetic disorder is diffuse, occurring throughout reciprocal space. For this reason, it is easily distinguished from the diffraction that arises from any ordered component, that is localised into sharp Bragg peaks. The diffuse scattering may nevertheless exhibit considerable detail, which can be analysed to reveal the structural properties of the magnetic disorder, as shown below.

3.2.13 Validity of the Static Approximation

The preceding sections show how equilibrium thermodynamic properties (which are necessarily time independent) may be directly obtained by neutron scattering under the assumption that energy transfers to and from the sample are negligible.

In the case that they are not negligible the sample is generally not in a true state of equilibrium, but rather in a uniform steady state, close to equilibrium, in which the neutron current continuously exchanges energy with the sample. Integration over frequencies in (3.6) still gives the true instantaneous correlation function for the sample in this steady state, but that is no longer the desired correlation function of the equilibrium state that would be obtained in the absence of the neutrons current. Application of the fluctuation dissipation theorem and detailed balance establishes the following relation [14]:

$$S^{\alpha\beta}(\mathbf{Q}, \omega)_{\text{diffuse}} = \frac{kT\chi^{\alpha\beta}(\mathbf{Q})}{g^2\mu_B^2} \frac{\hbar\omega/kT}{1 - \exp(-\hbar\omega/kT)} F^{\alpha\beta}(\mathbf{Q}, \omega), \quad (3.14)$$

where F , called the spectral shape function, is normalised to unit area when integrated over ω . If $\alpha = \beta$ or if $\mathbf{q}$ is equivalent to $-\mathbf{q}$ it is an even function of ω .

Thus, to get a true equilibrium correlation function or susceptibility the integral in (3.6) should be corrected by a frequency and temperature dependent factor:

$$kT\chi^{\alpha\beta}(\mathbf{Q}) = \int_{-\infty}^{+\infty} \frac{1 - \exp(-\hbar\omega/kT)}{\hbar\omega/kT} S^{\alpha\beta}(\mathbf{Q}, \omega) d\omega. \quad (3.15)$$

The actual integration can either be performed explicitly on the estimated $S(\mathbf{Q}, \omega)$ or automatically, by using a large enough incident wavevector so that the spread of scattered wavevectors is within the instrumental resolution.⁸ The latter method is usually preferable.

The diffuse scattering at zero energy transfer measures $\sim T\chi(\mathbf{Q})F(\mathbf{Q}, 0)$ which is not equal to the equilibrium $T\chi(\mathbf{Q})$. While it may occasionally be expedient to consider this measured quantity (or more generally $\sim T\chi(\mathbf{Q})F(\mathbf{Q}, \omega_0)$ where ω_0 is some fixed frequency), it should be born in mind that it is only an approximation to the wavevector-dependent susceptibility. In general, the validity of the static approximation should always be carefully considered when discussing experimental data.

3.2.14 Generalised Susceptibility

In order to compare the results of neutron scattering with those of other experimental techniques (particularly *ac*-susceptibility), it is useful to adopt the ‘common language’ of generalised susceptibility. The dynamic scattering function $S^{\alpha\beta}(\mathbf{Q}, \omega)_{\text{diffuse}}$ may be alternatively expressed in terms of an complex frequency-dependent susceptibility $\chi(\mathbf{Q}, \omega) \equiv \text{Re}[\chi(\mathbf{Q}, \omega)] + i\text{Im}[\chi(\mathbf{Q}, \omega)]$. Here, the real

⁸ Note that $\mathbf{Q}$ scales linearly with k , while energy scales quadratically, which means that the spectrum effectively narrows with increasing $\mathbf{k}_i$.

and imaginary parts mediate, respectively, the in phase and out of phase response of the system to a time dependent magnetic field. Bragg scattering, as it reflects a broken ergodicity in the system, does not depend on time or frequency. The diffuse scattering, in contrast, usually will depend on time, as it reflects the disordered part of the spin system. The fluctuation-dissipation theorem connects the dissipation of neutron energy to the spectrum of spontaneous fluctuations of the sample bathed in the neutron beam. The following relation can be established [14]:

$$S^{\alpha\alpha}(\mathbf{Q}, \pm|\omega|)_{\text{diffuse}} \propto \left(n_{|\omega|} + \frac{1}{2} \pm \frac{1}{2} \right) \text{Im}[\chi(\mathbf{Q}, |\omega|)], \quad (3.16)$$

where $n_{|\omega|} = (e^{\hbar|\omega|/kT} - 1)^{-1}$ may be interpreted as a harmonic oscillator occupation number. If neutrons accept energy from the sample (neutron energy gain), what is measured is essentially the thermal spectrum, while if neutrons give energy to the sample (neutron energy loss), the scattering measures states excited by the neutrons, which need not be thermally populated. The two cross sections differ by a factor of $e^{-\hbar|\omega|/kT}$, an expression of the condition of ‘detailed balance’.

3.2.15 Neutron Spectroscopy

In the case that the sample contains discrete excitations, these correspond to poles in the dynamical susceptibility and it is appropriate to adopt a more ‘spectroscopic’ description, involving matrix elements and selection rules. Perhaps the most basic excitations in ionic magnetic solids are so-called “magnetic excitons”: dispersed single ion excitations with (possibly gapped) dispersion $\omega^i(\mathbf{q})$ [15]. Here, $i = 0, 1, 2, \dots$ represent the ladder of corresponding single ion excitations. The lowest energy excitations ($i = 0$) are the magnons while higher excitons originate in the single ion crystal field states. If exchange is small compared to the crystal field splitting, as is typical in rare earth salts, then the dispersion of the magnetic excitons may be neglected and neutron scattering can be used to estimate the single ion crystal field spectrum. The unpolarised cross section for a particular transition $i \rightarrow j$ is

$$\frac{d^2\sigma}{d\Omega dE} = r_0^2 \frac{k_f}{k_i} F(|\mathbf{Q}|)^2 n_i |\langle j | \hat{\mu}_\perp | i \rangle|^2 \delta(E_i - E_j + E) \quad (3.17)$$

(neglecting Debye–Waller factor) where n_i is the number of ions in the i th state, of energy E_i , and $\hat{\mu}_\perp$ is the component of the magnetic moment operator perpendicular to the scattering vector. It can be seen from this expression that the intensities of the transitions depend in a very sensitive way on the single ion wavefunctions, so neutron scattering is a direct and powerful method of estimating these functions and deriving the crystal field parameters. However, it should be noted that powder averaging gives potential ambiguity in the assignment of crystal field transitions, so results obtained from powder samples should always be checked for consistency with other measurements, for example, bulk susceptibility or heat capacity.

3.3 Typical Neutron Scattering Patterns

In the previous section, I have tried to illustrate some of the different interpretations of neutron scattering. Thus, in the static approximation neutron scattering may be regarded as a thermodynamic probe that reveals properties the sample's equilibrium state, independent of the neutron flux. Beyond the static approximation it can either be regarded as probing the sample's relaxation towards that equilibrium state, following the weak disturbance of the neutron flux, or as a spectroscopy that reveals the sample's energy level structure, within the usual caveat of selection rules.

In this section, I aim to illustrate how these various probes of the sample may be used to classify magnetic states, and in particular, to discriminate conventional magnets from highly frustrated magnets. To this end I consider some typical neutron scattering patterns.

3.3.1 Scattering Plane

In a typical neutron scattering experiment, the scattering triangle is in a horizontal plane called the *scattering plane*. As illustrated in Fig. 3.5, we may adopt a set of cartesian axes in which z is normal to the scattering plane (and so perpendicular to $\mathbf{q}$) and x and y are within the scattering plane. The unpolarised experiment (3.7 and 3.11) measures

$$\chi^{zz}(\mathbf{q}) + (1 - \hat{q}_x^2)\chi^{xx}(\mathbf{q}) + (1 - \hat{q}_y^2)\chi^{yy}(\mathbf{q}) - 2\hat{q}_x\hat{q}_y\chi^{xy}(\mathbf{q}). \quad (3.18)$$

where $\hat{q}_x^2 + \hat{q}_y^2 = 1$. Consider, for example, a conventional cubic paramagnet with $z = [001]$, in which case $\chi^{xx}(\mathbf{q}) = \chi^{yy}(\mathbf{q}) = \chi^{zz}(\mathbf{q}) = \chi(\mathbf{q})$; $\chi^{xy} = 0$. The above expression simply becomes $2\chi(\mathbf{q})$ so the wavevector dependent susceptibility is directly measured.

A large number of neutron scattering experiments on cubic frustrated magnets (spinel or pyrochlores) have typically investigated the $[1\bar{1}0]$ scattering plane. This plane, with two-fold symmetry, contains the major symmetry directions $[001]$, $[110]$ and $[111]$. Its reciprocal lattice, for face centred cubic symmetry, is illustrated in Fig. 3.6 [16].

3.3.2 Free Energy

In order to evaluate the wavevector dependent susceptibility, we consider the Helmholtz free energy, which is the thermodynamic potential to be minimised when the magnetic moment M is constrained to a certain value: although this can't be achieved in practice we can use it conceptually to discuss the susceptibility. At a fixed temperature, the magnetic field H is the first derivative of the free energy with

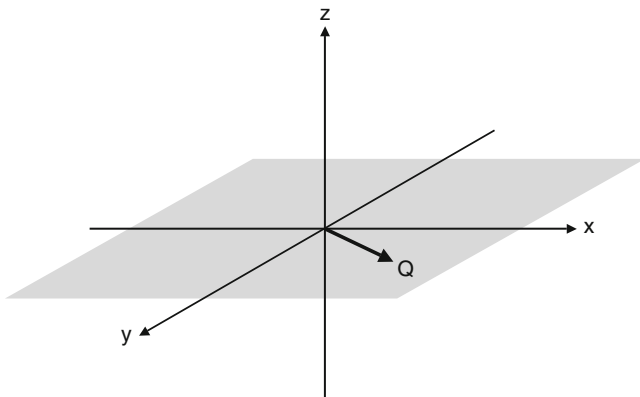


Fig. 3.5 Illustration of a horizontal scattering plane (*grey*) and axis system in a conventional single crystal scattering experiment

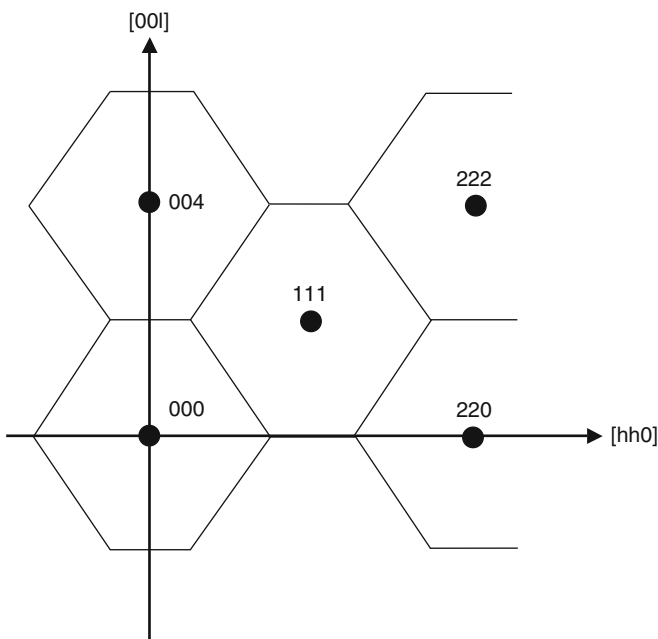


Fig. 3.6 Reciprocal lattice for the *hhl* plane in face centred cubic symmetry

respect to magnetic moment, $H = (\partial F / \partial M)_T$, and the inverse susceptibility is the second derivative: $\chi^{-1} = (\partial^2 F / \partial M^2)_T$ [17]. As an example, consider the case of an ideal paramagnet. If we write $F(M) = F_0 + \frac{1}{2}\chi_0^{-1}M^2 - MH$ (which assumes that the magnetic moment and field vectors are parallel) then a first differentiation shows that the free energy is minimised at $M = \chi_0 H$, which is indeed a property of a paramagnet, and a second differentiation confirms that the susceptibility

χ_0 is consistently defined (here F_0 is the component of F that does not depend on M). The quadratic dependence of F on M shows that the paramagnet resists magnetization in the absence of an applied field. This resistance is purely statistical or entropic as there is no energetic penalty in magnetizing a paramagnet: it doesn't magnetise spontaneously simply because there are so many more states with zero magnetic moment than there are states with finite magnetic moment. Because the internal energy U for the paramagnet is zero, the above free energy expression can alternatively be written $F = -TS - MH$. The susceptibility is given by the Curie Law: $\chi_0 = C/T$.

We can generalise this approach to predict the form of the neutron scattering in various types of magnets as we now show. The basic idea is to write down a similar free energy expression for a spatially varying magnetic moment field $\mathbf{M}(\mathbf{r})$ transformed into reciprocal space.

3.3.3 Ideal Paramagnet

We first consider the case of the ideal paramagnet and note that, as well as being resistant to magnetization, it is equally resistant to ordering at any wavevector $\mathbf{q}$ so we should be able to write:

$$F - F_0 = \frac{1}{2} \chi_0^{-1} \sum_{\mathbf{q}} \mathbf{M}(\mathbf{q}) \cdot \mathbf{M}(-\mathbf{q}) - M_x(\mathbf{q}) H_x(-\mathbf{q}) \quad (3.19)$$

where the field direction defines the x -axis. Minimising this expression with respect to each component of $\mathbf{M}(\mathbf{q})$ individually gives $\chi^{\alpha\alpha}(\mathbf{q}) = H^\alpha(\mathbf{q})/M^\alpha(\mathbf{q}) = \chi_0$, with terms in $\alpha\beta$ equal to zero. Then from (3.7, 3.11) we see that the diffuse neutron scattering does not vary with temperature and, apart from the slow variation due to magnetic form factor, is constant throughout reciprocal space. This is a scattering pattern of the type anticipated in Fig. 3.1a.

3.3.4 Conventional Magnet Above T_C

We next consider a conventional magnet in its paramagnetic phase [14]. There is a tendency to order at some wavevector $\mathbf{k}$ (see Sect. 3.2.11), but the statistical resistance to ordering is sufficient to overcome that tendency. For simplicity we consider the case of a ferromagnet with $\mathbf{k} = 0$. We have to add a term to our free energy expression that represents the internal energy arising from the magnetic exchange coupling. We imagine a mean magnetic moment field surrounding a given lattice site $\mathbf{r}$ that is just slightly smaller than the magnetic moment field at the lattice site itself. The mean field may be represented as a Taylor expansion: $\mathbf{M}_{MF} \propto \mathbf{M}(\mathbf{r}) + (a^2/2)\nabla^2\mathbf{M}(\mathbf{r})$ where a is the lattice constant. Summing over z nearest neighbours the total exchange energy is $-zJ \int \mathbf{M}(\mathbf{r}) \cdot (\mathbf{M}(\mathbf{r}) + (a^2/2)\nabla^2\mathbf{M}(\mathbf{r})) d\mathbf{r}$,

where J is the exchange constant. By substituting (3.8) into this expression, and integrating, we find that the internal energy of the magnet, can be represented in reciprocal space by the expression $U = zJ \sum_{\mathbf{q}} \mathbf{M}(\mathbf{q}) \cdot \mathbf{M}(-\mathbf{q})((a^2/2)q^2 - 1)$. Including the entropic factor and field energy (3.19), the free energy becomes:

$$F - F_0 = \frac{1}{2} \sum_{\mathbf{q}} \mathbf{M}(\mathbf{q}) \cdot \mathbf{M}(-\mathbf{q}) (\chi_0^{-1} - zJ + zJa^2q^2) - \mathbf{H}(\mathbf{q}) \cdot \mathbf{M}(-\mathbf{q}). \quad (3.20)$$

Differentiating, and forming $\chi^{\alpha\beta} = dM^\alpha/dH^\beta$ we find the following expression for the wavevector dependent susceptibility:

$$\chi^{\alpha\alpha}(\mathbf{q}) = \chi(q) = \frac{\chi(0)\kappa^2}{q^2 + \kappa^2} \quad (3.21)$$

with $\chi^{\alpha\beta} = 0$ for $\alpha \neq \beta$. In (3.21) the parameters $\chi(0)$ and κ are defined by the two relationships $\chi(0) = C/(T - zJC)$, which we recognise as the Curie–Weiss expression, and $\chi(0) = (a\kappa)^{-2}$, a scaling law.⁹ Here, we have assumed a cubic sample (in other cases both χ and κ would depend on $\alpha = x, y, z$), and used $\chi_0 = C/T$. Equation (3.21) leads to the famous Ornstein–Zernike form of the scattering function $S(\mathbf{Q}) \sim T\chi(\mathbf{q})$. It may be Fourier transformed to give a correlation function $g(r) = r^{-1} \exp(-\kappa r)$, so it can be seen that $\xi = 1/\kappa$ measures the size of the correlated regions, i.e., ξ is the correlation length. In reciprocal space, the scattering forms diffuse blobs of spherical symmetry centred on the ferromagnetic Brillouin zone centre with width $\sim \kappa$, the form anticipated in Fig. 3.2b (in a non cubic magnet these blobs will generally have ellipsoidal symmetry).

3.3.5 Conventional Magnet Below T_C

In this case, following (3.11) the neutron scattering is a superposition of (3.21) (Fig. 3.1b) with a sharp Bragg peak at the Brillouin zone centre. Just below T_C , the intensity of the Bragg peak grows as the square of the magnetization, i.e., as $\sim (T_C - T)^{2\beta}$, where β is the order parameter critical exponent. The diffuse scattering is much weaker than the Bragg scattering apart from very close to the critical temperature, where it is approximately described by (3.21), with $\chi(0) \sim |T - T_C|^{-\gamma}$ and $\kappa \sim |T - T_C|^\nu$: here γ and ν are the critical exponents for susceptibility and correlation length respectively.

⁹ This is a mean field expression: in critical point theory the susceptibility acquires an ‘anomalous’ dimension $\chi(0) \sim \kappa^{-2+\eta}$ [18].

3.3.6 Cooperative Paramagnet

As a prototypical case, we consider spin ice, the highly frustrated pyrochlore ferromagnet [16, 19]. Its low temperature cooperative paramagnetic phase is governed by the ice rule constraint ‘two spins in, two spins out’ per tetrahedron of the pyrochlore structure. The magnetic moment field $\mathbf{M}(\mathbf{r})$ is therefore solenoidal (i.e., a solenoid has as many magnetic flux lines flowing into it as out of it) and obeys the local ‘divergence free’ condition:

$$\nabla \cdot \mathbf{M}(\mathbf{r}) = 0, \quad (3.22)$$

from which follows

$$\mathbf{q} \cdot \mathbf{M}(\mathbf{q}) = 0 \quad (3.23)$$

and the system is thus completely described by Fourier components for which magnetic moment is perpendicular to its wavevector $\mathbf{q}$. As an approximation, which ignores the details of the local spin structure, we may consider spin ice to be a paramagnet in which the free energy (3.19) is minimised subject to this constraint, so as before, we seek to calculate $\chi(\mathbf{q})$ by differentiating the free energy. In three dimensions, the algebra is laborious, but in two dimensions it is very easy [20, 21]. The constraint (3.23) may be written

$$q_x M_x(\mathbf{q}) + q_y M_y(\mathbf{q}) = 0, \quad (3.24)$$

so the magnetization components are not independent. First, consider the response of M_x to a field along x , so $\mathbf{H} = (H_x, 0, 0)$. The free energy is:

$$F = \sum_{\mathbf{q}} \frac{1}{2} \chi_0^{-1} (M_x^2(\mathbf{q}) + M_y^2(\mathbf{q})) - M_x(\mathbf{q}) H_x(\mathbf{q}), \quad (3.25)$$

so by eliminating M_y , using (3.24), and by minimising with respect to M_x as above, we find:

$$\chi^{xx} = \chi_0 \frac{q_y^2}{q_x^2 + q_y^2}. \quad (3.26)$$

We could have alternatively considered the response of M_x to a field along y , with the result:

$$\chi^{xy} = -\chi_0 \frac{q_x q_y}{q_x^2 + q_y^2}. \quad (3.27)$$

The neutron scattering function $S^{\alpha\alpha}(\mathbf{q}) \sim \chi^{\alpha\alpha}(\mathbf{q})T$, is given by (3.26). This is an unusual functional form that contains a singularity as $q \rightarrow 0$: if the origin is approached along y at $q_x = 0$ then $\chi(0) = \chi_0$ whereas if it is approached along x at $q_y = 0$ then $\chi(0) = 0$. The scattering function is therefore singular at the origin: a so-called pinch point singularity or ‘bow tie’ of scattering, as anticipated in Fig. 3.2c. Recalling that we ignored the details of the local spin structure, this form of the scattering would not be expected to apply at large wavevector in the Brillouin zone, but it should be a good approximation near the zone centre.

The pinch point scattering ($\sim \chi T$) may be re-expressed in polar coordinates $q = \sqrt{q_x^2 + q_y^2}$, $\theta = \arcsin(q_x/q)$, showing that the diffuse scattering near the zone centre depends only on angle θ , not on q (see Fig. 3.2c). Alternatively one can write (3.26), (3.27) as

$$\chi^{\alpha\beta} = \chi_0 (\delta_{\alpha,\beta} - \hat{q}_\alpha \hat{q}_\beta), \quad (3.28)$$

where $\hat{q}_{\alpha,\beta}$ are unit vector components.

This analysis is easily extended to three dimensions. Taking into account the summation over cartesian components in (3.7) pinch points may not be visible to unpolarised neutrons in a four-fold scattering plane like [100], but should be visible in a two fold scattering plane like [110]; in either case it is clear that they would be better studied by polarised neutron scattering which can isolate the components of $\chi^{\alpha\beta}$.

Pinch point scattering is not only a property of spin ice but also applies in principal to any highly frustrated magnet that obeys a local ice-rule type constraint [22, 23]. The scattering is seen to be fundamentally different to that of a simple paramagnet or a disordered ferro- or antiferromagnet. When (3.28) is Fourier transformed, the spin correlations in direct space are seen to take the form of a dipolar interaction, with a complicated angular dependence and a slow algebraic decay as the inverse cube of distance [20]. This contrasts the decay of correlations in a conventional magnet above its ordering temperature, where the decay of correlations is like a screened Coulomb interaction (3.21). This relationship of correlation functions to charge interactions was first pointed out by Youngblood, Axe and McCoy [21] for the analogous case of ice-rule paraelectrics. The pinch point scattering functions of cubic magnetic systems have been analysed in detail by Isakov et al. [22] and by Henley [23]. Their relevance to frustrated magnets was first realised by Zink et al., on the basis of numerical simulations [24], and the first clear experimental observation was recently reported by Fennell et al. [25].

Comparing Figs. 3.2a and 3.1b we finally see how neutron scattering reveals a symmetry difference, in the spin correlation function, between a cooperative paramagnet and a conventional paramagnet. The cooperative paramagnetic phase may be called a ‘Coulomb phase’, by analogy with field theories, because its basic excitations can be shown to be effective charges or magnetic monopoles [26].

3.3.7 Absent Pinch Points

In any real frustrated magnet, it is certain that the pinch point singularity will be removed by one mechanism or another, even though this effect may be very small. A first possibility is that the zero divergence constraint (3.22) is relaxed at a long length scale ξ_{ice} by adding a term in $(\nabla \cdot \mathbf{M}(\mathbf{r}))^2$ to the free energy [20]. This will cause a broadening of the pinch point to give it a width ξ_{ice}^{-1} (see Fig. 3.7, left), and the susceptibility takes the form:

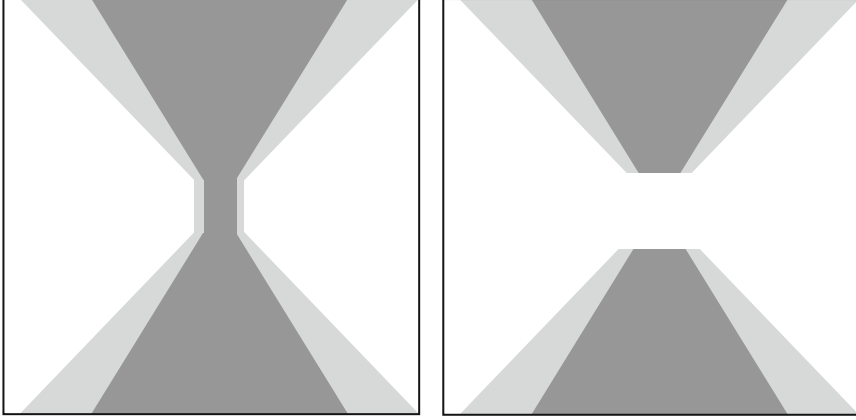


Fig. 3.7 Absent pinch points (schematic). *Left*: the pinch point is replaced by a ridge of width ξ_{ice}^{-1} . *Right*: the scattering near the zone centre is completely suppressed, indicating finite spin clusters (see text)

$$\chi^{xx} = \chi_0 \frac{q_y^2 + \xi_{\text{ice}}^{-2}}{q_x^2 + q_y^2 + \xi_{\text{ice}}^{-2}}. \quad (3.29)$$

A second possibility arises when there is an additional length scale in the problem that removes the power law correlations associated with the pinch point. This is most easily seen in the case of spin ice, where the spin ice state can be constructed from spin loops of all sizes, from the smallest possible, a hexagon, to loops that span the entire sample. Each of these loops is closed and carries zero magnetization: the algebraic nature of the correlations arises from the fact that there are loops on all scales. It is possible, however, to construct a state that consists *only* of hexagonal loops [27]. It is impossible to construct a long wavelength magnetization fluctuation from these finite loops so the scattering function is completely suppressed in the Brillouin zone centre (Fig. 3.7, *right*). In reality, unless there is some breaking of spin symmetry (typically signaled by a phase transition), then a complete suppression of the pinch point is unlikely.

A similar argument may be applied to frustrated antiferromagnets: a pattern like Fig. 3.7b would be indicative of a tendency to finite size clusters of some sort. In fact, experiments on systems such as $\text{Y}(\text{Sc})\text{Mn}_2$ [28], ZnCr_2O_4 [29] and MgCr_2O_4 [30] reveal that this is the common situation in pyrochlore lattice spin liquids (see Sect. 3.4).

3.3.8 Dynamical Signature of Cooperative Paramagnetism

It would be wrong to speak of a single dynamical signature of cooperative paramagnetism, as the spin dynamics depend in great detail on the actual interactions in the

system. For example, in Ising-like systems such as spin ice, the dynamics are thermally activated [26, 31] while in systems with continuous spin symmetry, they may persist to the limit of low temperature. Nevertheless, two aspects of spin dynamics in highly frustrated magnets could be regarded as generic.

First, for the classical Heisenberg antiferromagnet on the pyrochlore lattice, Moessner and Chalker derived the following form for the time dependence of the spin autocorrelation function [32]:

$$\langle \mathbf{S}_r(0) \mathbf{S}_r(t) \rangle = \exp(-cTt), \quad (3.30)$$

where c is a constant. This Fourier transforms to a Lorentzian scattering function in frequency space with width $\sim T$. A relaxation of the type (3.30) might be expected to be quite typical for highly frustrated magnets as it reflects a very large degeneracy of ground states and low lying excited states, that include dynamical modes of zero energy. Thus, even the smallest temperatures mediate an extremely rapid relaxation of any disturbance, and it is only at zero temperature that all motions completely disappear.

The second aspect of spin dynamics expected to be generic for highly frustrated magnets is the elementary thermal excitation of the Coulomb phase, such as the magnetic monopoles of spin ice [26]. These cost energy but do not couple directly to neutrons as their creation or destruction does not involve a single unit of spin angular momentum. Thus they could only be detected indirectly by their effect on the thermal or spatial evolution of the spin correlation functions.

3.4 Experimental Results

In this section, I describe various results of key neutron scattering experiments on highly frustrated magnets. This is not a comprehensive review of the literature, but rather, examples are chosen to illustrate the general ways by which neutron scattering may contribute to the field of highly frustrated magnetism.

3.4.1 Cooperative Paramagnet States

3.4.1.1 Spin Ice

Spin ice is the Ising-like highly frustrated pyrochlore lattice ferromagnet, in which spin configurations map onto proton configurations in Pauling's model of water ice [16, 19]. $\text{Ho}_2\text{Ti}_2\text{O}_7$, the canonical spin ice material [16, 33], is an ideal subject for neutron scattering, as it freezes into a disordered magnetic state below ~ 2 K, in which the static approximation is almost ideally obeyed. Thus, mapping of the diffuse scattering in reciprocal space gives a direct measure of $\chi^{\alpha\beta}(\mathbf{q})$. Furthermore,

the moments on Ho^{3+} are about the largest of any magnetic substance ($10\mu_B$), and since the neutron cross section scales as the square of the moment, its magnetic scattering is orders of magnitude stronger than that of a typical magnet. This is also true of another spin ice material, $\text{Dy}_2\text{Ti}_2\text{O}_7$ [34], but in other respects latter is less suited to neutron scattering as dysprosium is strongly neutron absorbing. To compensate for this it has been studied as the isotopically enriched $^{162}\text{Dy}_2\text{Ti}_2\text{O}_7$ [35].

The (unpolarised) scattering pattern of spin ice materials has been extensively mapped in several planes of reciprocal space in both zero and applied magnetic field [16,25,33,35–38]. In zero applied field the narrowing of the diffuse unpolarised scattering towards the Brillouin zone centre suggests the pinch points that are characteristic of pseudo-dipolar correlations, but it is only very recently that these have been clearly revealed by polarised neutron scattering [39]. Application of a magnetic field on [111] gives the reduced dimensionality kagomé ice phase [40]: this shows very clear pinch points (Fig. 3.8) [25]. The existence of pinch points shows that there is no characteristic length scale for the disorder in kagomé ice, so it is very different to a conventional paramagnet.

Away from the Brillouin zone centre the scattering pattern reflects the fine details of the spin–spin interactions in spin ice. The theory is well developed and a particularly accurate spin Hamiltonian has been established for $\text{Dy}_2\text{Ti}_2\text{O}_7$ by fitting the experimental $S(\mathbf{Q})$ to a parameterised theoretical form [27].

At temperatures above about 0.3 K, the spin relaxation of spin ice comes into the experimental window. It has been studied by the technique of neutron spin echo,

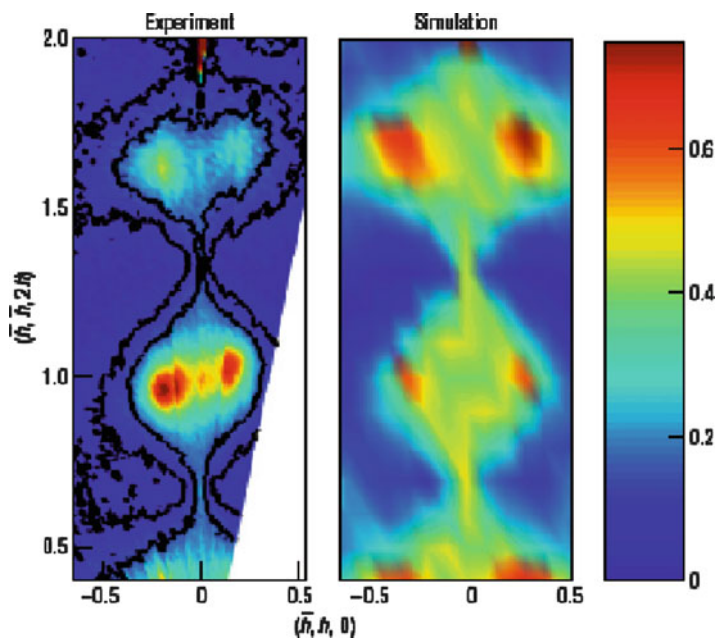


Fig. 3.8 Pinch point scattering in kagomé ice: experiment (*left*) versus theory (*right*) [25]

which measures $S(\mathbf{Q}, t)$ (called the intermediate scattering function), the Fourier transform of $S(\mathbf{Q}, \omega)$ [41, 42]. This revealed an athermal regime of constant relaxation, stretching from 2 to 15 K. Such behaviour is consistent with the diffusion of monopole defects [31]. At higher temperatures the same study revealed that relaxation occurs via single ion excitations to crystal field states. It should be noted that the spatial characteristics of neutron scattering were key to establishing this fact [41].

3.4.1.2 Pyrochlore and Spinel Antiferromagnets

The classical Heisenberg antiferromagnet on the pyrochlore lattice is one of the most basic spin models to exhibit a cooperative paramagnetic, or spin liquid state [32]. The system has a local ordering principle of zero magnetization per tetrahedron of the pyrochlore structure, that leads, as in the case of the ice rules, to pinch point singularities at Brillouin zone centres [22, 23]. However, as discussed in [43], the pure Heisenberg model is unphysical in the sense that it implies a spin symmetry that cannot exist in real pyrochlore materials. The inclusion of a realistic Ising or XY spin symmetry eliminates the spin liquid state in favour of ordered states, and, surprisingly, it is only the Ising *ferromagnet* (spin ice), rather than the antiferromagnet, that exhibits a macroscopic ground state degeneracy.

Nevertheless, there are several pyrochlore antiferromagnets with peculiarities that stabilise cooperative paramagnetic states. The one that perhaps comes closest to being a true Heisenberg antiferromagnet is the spinel ZnCr_2O_4 : at low temperatures (12 K) this system has an ordering transition with a lattice distortion, but above that temperature it approximates the Heisenberg antiferromagnet on an undistorted pyrochlore lattice. Lee et al. reported neutron scattering maps of the hhl plane at finite energy transfer: essentially maps of $S(\mathbf{Q}, \omega_0)$ at a finite ω_0 [29]. This function can be interpreted as roughly approximating the equilibrium spin correlations, but with some weighting towards those configurations preferentially excited by the neutron flux.¹⁰ Pinch points were not clearly observed: rather the data appeared to show diffuse blobs of scattering, reminiscent of the expected pattern, but separated by broad saddle points rather than by sharp pinch points. From the analysis of Sect. 3.3, this can be interpreted as evidence for a characteristic length scale in the disordered state. Indeed, Lee et al. showed that the scattering pattern can be well fitted by a model of hexagonal, non interacting clusters (‘emergent excitations’) (Fig. 3.9). The neutron scattering pattern bears a similarity to that predicted for the quantum Heisenberg antiferromagnet [44]. More recently, similar scattering has been observed in the related spinel MgCr_2O_4 [30].

Another material of great interest is the metallic alloy $(\text{Y}_{0.97}\text{Sc}_{0.03})\text{Mn}_2$, in which the Mn atoms occupy a pyrochlore lattice. The related phase YMn_2 is

¹⁰ Measuring at finite energy transfer typically ‘cleans up’ the pattern, by removing nuclear elastic scattering.

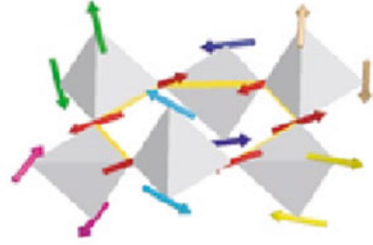
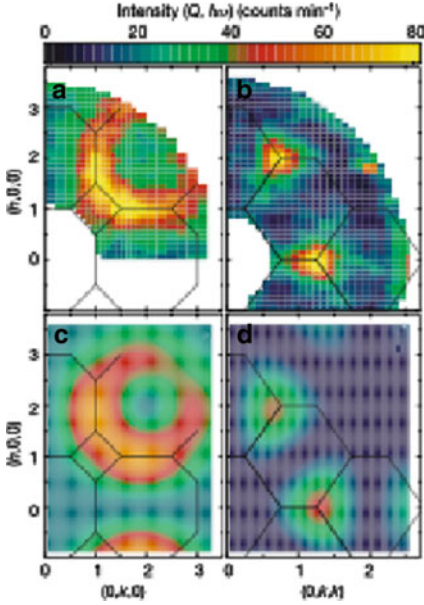


Fig. 3.9 Cluster like scattering in ZnCr_2O_4 [29]. *Left: experiment (top) versus theory (bottom); right: a six spin cluster*

antiferromagnetically ordered below 100 K, but the application of pressure or ‘chemical pressure’ (the substitution of some Sc) leads to an itinerant state with heavy Fermion behaviour. Neutron inelastic scattering shows that the magnetism in this phase is almost entirely dynamic in nature: localised spin fluctuations out of an itinerant state. Ballou et al. mapped the neutron scattering in the hhl plane at finite energy transfer. The scattering is strikingly similar to that of ZnCr_2O_4 , described above, as well as the predicted form for the quantum Heisenberg antiferromagnet [44].

A third Heisenberg-like antiferromagnet to be studied in detail is CsMnCrF_6 , in which Cr and Mn are distributed over a pyrochlore lattice [45]. Here, the Cr^{3+} and Mn^{2+} ions have only weak anisotropy, which is further frustrated by the positional disorder of the ions (Banks et al., unpublished). Neutron scattering is consistent with the expected pattern for the classical Heisenberg model, and broadened pinch points have recently been identified. The temperature evolution of the dynamic scattering function is broadly consistent with the expression (3.30) (see Fig. 3.10).

Finally, in this section, we mention the case of $\text{Tb}_2\text{Ti}_2\text{O}_7$ which has a spin liquid state at Kelvin temperatures. The diffuse scattering again shows broad features with no clearly resolved pinch points [46]. The degeneracy of the spin liquid can be lifted by the application of pressure, which induces a magnetic ordering transition [47].

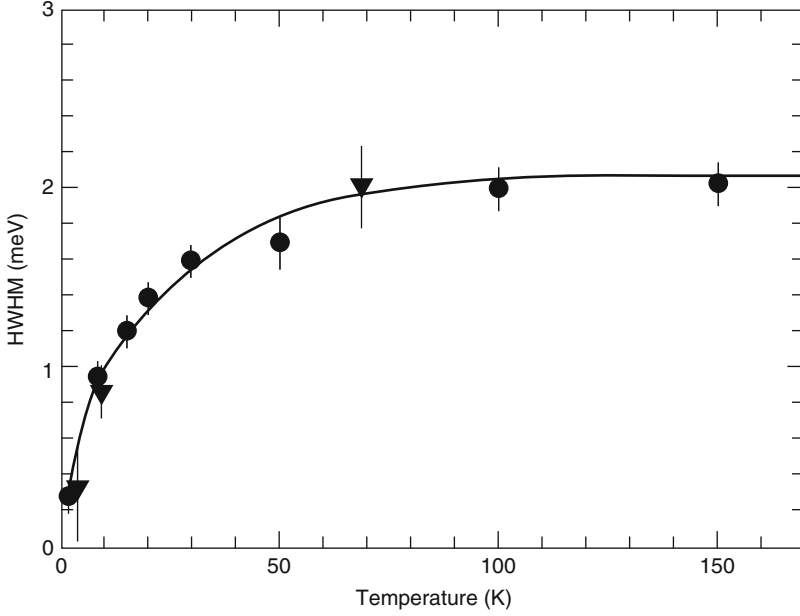


Fig. 3.10 Width of the inelastic scattering peak in CsMnCrF_6 as a function of temperature, showing it to narrow as $T \rightarrow 0$ [45]

3.4.1.3 Powder Experiments

The effect of powder averaging is to reduce the four-dimensional scattering function $S(\mathbf{Q}, \omega)$ into a two-dimensional one, $S(Q, \omega)$. This is a major problem in the study of cooperative paramagnets which are strongly characterised by the angular dependence of the scattering function (see Sect. 3.3). Nevertheless, the powder averaged scattering function does give some spatial information about the spin correlations. In the static approximation, powder averaging of $S(\mathbf{Q})$ gives

$$S(Q) = \sum_r \langle \mathbf{S}_0 \cdot \mathbf{S}_r \rangle z_r \sin(Qr)/(Qr), \quad (3.31)$$

where z_r is the number of spins at distance r . Using this expression, models for the spin correlation function can be tested against neutron scattering data. This technique has been applied to the study of spin correlations in many spin disordered phases: for example, the pyrochlores $\text{Y}_2\text{Mo}_2\text{O}_7$, $\text{Tb}_2\text{Ti}_2\text{O}_7$ and $\text{Tb}_2\text{Mo}_2\text{O}_7$ [48], the kagomé system deuterium iron jarosite [49], gadolinium gallium garnet [50], and $\beta\text{-Mn}_{1-x}\text{Co}_x$ [51], as well as partially ordered $\text{Gd}_2\text{Ti}_2\text{O}_7$ [13] (see below).

3.4.2 Ordered States

The resolution of frustration that occurs in many highly frustrated magnets leads to magnetically ordered (or partly ordered) states. Although all spin ordered phases are, in a sense, conventional, their study may nevertheless reveal essential aspects of the physics of geometric frustration, as well as material properties that cannot be estimated from the study of disordered states alone. Thus, magnetic structure is a very sensitive measure of the interactions in the system, and the mechanism by which frustration is relieved. Neutron scattering is the most direct, robust and flexible method of magnetic structure determination, so plays a crucial role in such studies. The examples discussed in this section are particularly chosen to illustrate these points.

3.4.2.1 Ising Pyrochlore: FeF_3

FeF_3 may be considered to approximate the most basic of all realistic magnetic models on the pyrochlore lattice: the Heisenberg antiferromagnet with $\langle 111 \rangle$ Ising-like single ion anisotropy. As mentioned above, frustration is strongly relieved to give an ordered state with spins on alternate tetrahedra of the pyrochlore structure all pointing ‘in’ or ‘out’. This magnetic structure was confirmed by powder neutron scattering [52]. Despite being ordered, the low temperature state shows an interesting property that has been associated with the geometric frustration of the lattice. The temperature dependence of the magnetic Bragg peaks, and hence the magnetic order parameter, is unconventional. The magnetic order parameter exponent β has been estimated to take the value $\beta = 0.18$, as distinct to the usual $\beta \approx 1/3$ for a typical three-dimensional magnet. It would be interesting to measure the exponent with greater precision by neutron diffraction on a single crystal sample of FeF_3 . Unconventional exponents are also observed in triangular lattice frustrated systems [53], but it should be noted that the example of $\text{Er}_2\text{Ti}_2\text{O}_7$, discussed below, rules out the possibility that they are universal in ordered frustrated magnets.

3.4.2.2 XY Pyrochlore: $\text{Er}_2\text{Ti}_2\text{O}_7$

Erbium titanate approximates the Heisenberg pyrochlore antiferromagnet with local $\langle 111 \rangle$ XY-like single ion anisotropy [54]. The classical model relieves frustration with order induced by thermal fluctuations (‘order by disorder’) [54, 55]. The magnetic structure formed has a single tetrahedron basis with every spin lying along a local $\langle 211 \rangle$ axis, in the ‘XY’ plane. The real material orders at $T_N = 1.2$ K to form exactly this structure. There is no mechanism other than order by disorder known to stabilise it: in particular, the leading perturbation to the model Hamiltonian – dipolar interactions – favour a different structure, as discussed below. Thus, it was suggested in [54] that the magnetic structure is stabilised by quantum order by disorder that mimics the classical thermal order by disorder. Two caveats on this argument have

recently been removed. First, the existence of (previously unobserved) Goldstone mode spin waves has been demonstrated by neutron inelastic scattering [56], lending support to a Hamiltonian with continuous symmetry, like that proposed. Second, 6-fold crystal field anisotropy has been shown to be too weak to be the sole cause of the observed ordering pattern, again lending support to the simplified Hamiltonian [57]. In this context, further theoretical studies of the quantum XY pyrochlore will be of great interest to confirm if $\text{Er}_2\text{Ti}_2\text{O}_7$ really does have a ground state stabilised through the order by disorder mechanism.

$\text{Er}_2\text{Ti}_2\text{O}_7$ exhibits two other points of interest [54]. First, as mentioned above, the order parameter exponent β has been measured by single crystal neutron diffraction. It takes a conventional value, $\beta = 0.33$, which is close to the theoretical value for the three-dimensional XY antiferromagnet, thus ruling out the idea that unconventional exponents are *universal* for frustrated magnets. Second, neutron diffraction leads to an ambiguity in the magnetic structure as two different (symmetry related) candidate structures give identical Bragg intensities, when magnetic domains are accounted for. The resolution of this difficulty illustrates some general neutron diffraction methodology. In the earliest study a magnetic field was applied to the single crystal sample to stabilise a magnetic monodomain: changes in diffracted intensity clearly discriminated the candidate structures, but it could always be argued that the application of a field to a frustrated magnet might alter the magnetic state. Thus, a more recent study [58] used a sophisticated zero field neutron polarimetry method to distinguish the structures and estimate domain populations, which finally confirmed the magnetic structure proposed in [54].

3.4.2.3 Dipolar Ordering: $\text{Gd}_2\text{Sn}_2\text{O}_7$

Dipole–dipole interactions are inescapable in any real magnet, but are rarely sufficiently strong to dictate the ordering pattern. Heavy rare earth salts have particularly strong dipolar interactions and relatively weak exchange. Salts of Gd^{3+} furthermore have relatively weak (though non-negligible) crystal field anisotropy, so gadolinium pyrochlores may be taken as a prototypical Heisenberg-dipolar frustrated magnets. Early ideas that $\text{Gd}_2\text{Ti}_2\text{O}_7$ should be a model system of this sort were quickly dispelled by neutron diffraction of the isotopically enriched $^{160}\text{Gd}_2\text{Ti}_2\text{O}_7$ [59],¹¹ but more recently it has been shown by powder neutron diffraction [12] that its close relative (enriched) $\text{Gd}_2\text{Sn}_2\text{O}_7$ does exhibit the magnetic structure expected for dipole–dipole interactions on the pyrochlore lattice [60]. This structure has spins in local ‘XY’ planes along $\langle 110 \rangle$ axes. It is probable that the single ion anisotropy of Gd^{3+} ¹² favours local XY symmetry, so does not compete with the dipolar ordering pattern.

¹¹ The isotope was used because natural Gd is strongly absorbing.

¹² Although Gd^{3+} is nominally an S -state ion, it normally exhibits single ion g -factor anisotropy of magnitude several tenths of a Kelvin.

3.4.2.4 Partial Order: $\text{Gd}_2\text{Ti}_2\text{O}_7$

Originally considered a good candidate for the Heisenberg-dipolar pyrochlore anti-ferromagnet [60, 61], there have been many conjectures about the properties of $\text{Gd}_2\text{Ti}_2\text{O}_7$, but these have been mainly ruled out by neutron diffraction evidence. At the time of writing the magnetic structures it exhibits remain only poorly understood, but are, nevertheless, of considerable interest.

In zero applied field $\text{Gd}_2\text{Ti}_2\text{O}_7$ shows two ordering transitions at 1.1 K and 0.7 K. In the first transition at 1.1 K only three quarters of the spins of the pyrochlore structure order magnetically [13]. The structure is partially ordered, consisting of one disordered tetrahedron per unit cell, as illustrated in Fig. 3.11. It is instructive to note that this is a ‘4- k ’ structure that looks totally different to the corresponding single k variant (illustrated in [59]). As described in Sect. 3.2, simple neutron diffraction cannot distinguish the single k and multi- k variants, but a study of the diffuse scattering arising from the disordered spin set, clearly favours the 4- k solution (see Fig. 3.12) [13]. The 0.7 K ordering transition then corresponds to the ordering of these spins, although they show strong quantum mechanical fluctuation in the ground state.

The actual powder pattern for $\text{Gd}_2\text{Ti}_2\text{O}_7$ is illustrated in Fig. 3.13. For symmetry reasons the $(1/2, 1/2, 1/2)$ magnetic peak arises solely from the ordering of the spin set that is remains disordered between 0.7 and 1.1 K: the rapid growth of this peak below 0.7 K proves the ordering of this set and gives a strong experimental handle on the ground state moment.

3.4.2.5 Kagomé Lattice: Jarosite Materials

The Jarosite family, in which M^{3+} spins ($\text{M} = \text{V}, \text{Cr}, \text{Fe}$) occupy a kagomé lattice in well-separated layers, affords many chemical variations which generally show

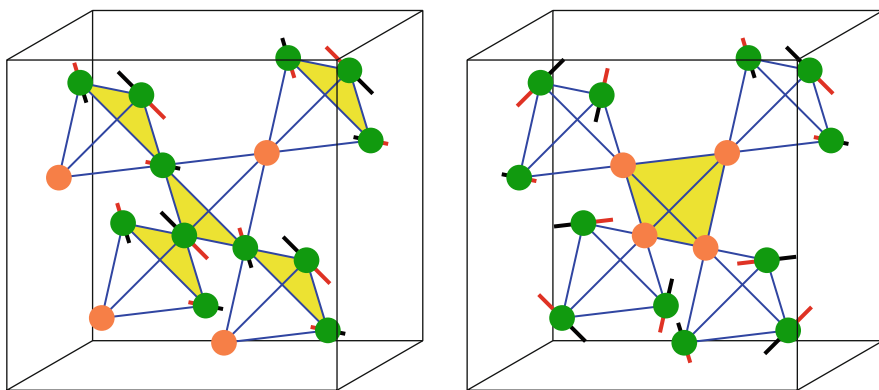


Fig. 3.11 Partially ordered magnetic structures of $\text{Gd}_2\text{Ti}_2\text{O}_7$ [13]. *Left*: the 1- k structure [59] and *right* the 4- k structure [13]. The 4- k structure is that found in experiment (see Fig. 3.12). The orange circles represent spins that are disordered above 0.7 K

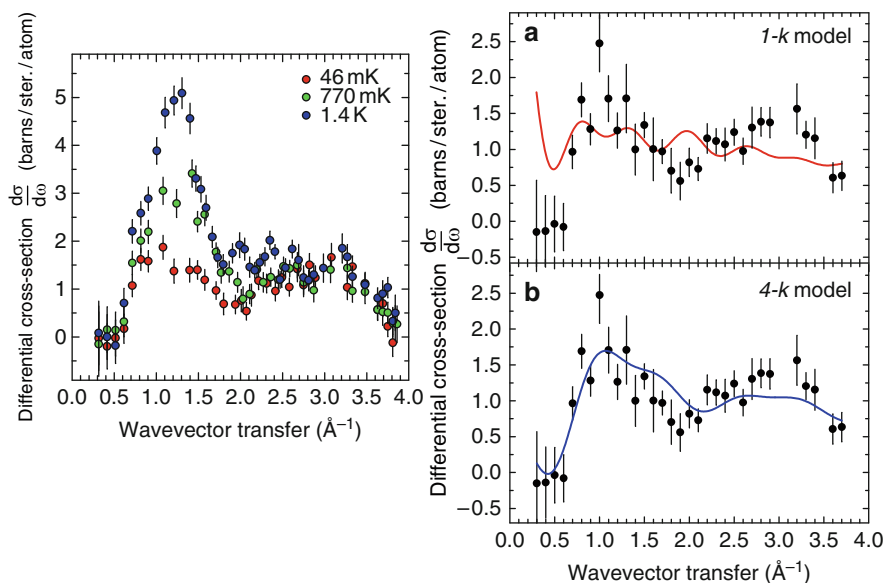


Fig. 3.12 Diffuse scattering from partly ordered $\text{Gd}_2\text{Ti}_2\text{O}_7$ [13]. The fits show that the disordered spins of the 4- k model describe the data much better than do those of the 1- k model (see Fig. 3.11)

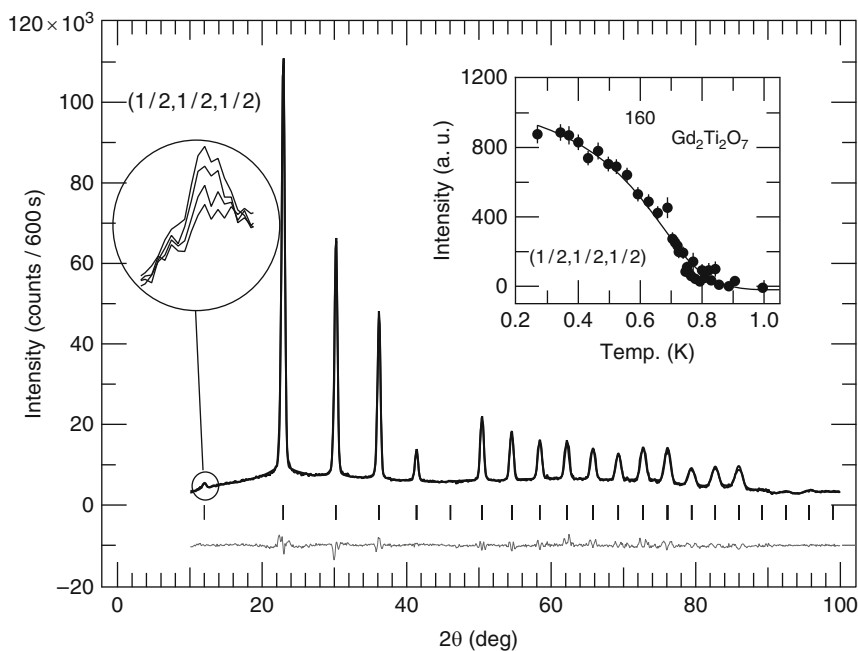


Fig. 3.13 Powder diffraction pattern of $\text{Gd}_2\text{Ti}_2\text{O}_7$ [13]. Lower line is observed data minus that calculated using the model of Fig. 3.11 and the insets show the growth of the small peak that arises exclusively from the spin set that is disordered above 0.7 K

magnetic ordering transitions (hydronium iron jarosite is the exception). Wills [62] has presented a representational analysis scheme for the structure which is used to derive possible magnetic structures that are consistent with an observed propagation vector $\mathbf{k}$ (see Sect. 3.2). For actual magnetic structure solutions on powder data see, for example, [62] and [63].

3.4.3 *Excited States*

The study of magnetic excited states by neutron scattering affords a definitive characterisation of the effective spin Hamiltonian of a ionic magnet. As described above, the basic excitations are magnetic excitons, those arising from the single ion ground level being the conventional spin waves, and others being best thought of as dispersed crystal field excitations. Key to the description of a magnetic solid (neglecting hyperfine effects) is the relative strength of the crystal field, the spin orbit coupling and the spin–spin interactions (exchange or dipolar). The relative values of these three energy scales determines the appropriate physical description. At a practical level the neutron scattering energy resolution is a fourth energy scale that, in comparison with the other three, determines what can be measured. The examples in this section are chosen to represent the various possibilities.

3.4.3.1 $\text{Ho}_2\text{Ti}_2\text{O}_7$: Crystal Field States

Spin ice $\text{Ho}_2\text{Ti}_2\text{O}_7$ represents the case where there is a very clean separation of energy scales in the system [16, 19]. Spin-orbit coupling is dominant, so that free Ho^{3+} ions are very well described in the basis $|J, m_J\rangle$: here $J = 8, m_J = 8, 7, 6 \dots -8$, with other J -levels being at much higher energy and hence negligible in a low energy description. The next strongest scale, the crystal electric field, lifts the degeneracy of these free ion states such that each level in the crystal field is a particular admixture of the $|J, m_J\rangle$'s consistent with the trigonal point symmetry of the pyrochlore structure. Exchange and dipolar interactions are much weaker again, so it is a good approximation to regard the magnetic excitons as undispersed crystal field levels. Neutron spectroscopy may then be used to measure the energy separations of these levels and the matrix elements between them. In [64], a spin Hamiltonian equivalent to the following was fitted:

$$H_{CF} = B_0^2 C_0^2 + B_0^4 C_0^4 + B_3^4 (C_3^4 - C_{-3}^4) + B_0^6 C_0^6 \\ + B_3^6 (C_3^6 + C_{-3}^6) + B_6^6 (C_6^6 + C_{-6}^6), \quad (3.32)$$

where the C s are tensor operators related to the Stevens operators. The derived crystal field parameters are given in [64]. Crystal field wavefunctions are a further product of the analysis: these show that the ground term is a doublet consisting almost entirely of $|8, \pm 8\rangle$, with the next excited state at ~ 300 K above the ground

state. As the dipolar coupling in $\text{Ho}_2\text{Ti}_2\text{O}_7$ is only of order ~ 1.5 K (exchange being weaker still) it is thus an extremely good approximation to regard the Ho^{3+} effective spins as Ising-like, the basis of spin ice physics.

Although the separation of energy scales in $\text{Ho}_2\text{Ti}_2\text{O}_7$ allows an extremely accurate characterisation of the spin Hamiltonian, this comes at a price. First, the weak coupling is not far off the limit of neutron resolution, meaning that excitations arising from the magnetic coupling cannot be easily resolved. Second, the Ising-like ‘spins’ do not have a true $S = 1/2$, so cannot be flipped by a neutron; thus the basic monopole excitations in spin ice cannot be excited or de-excited by the neutron flux, so are not directly visible to neutron spectroscopy.

3.4.3.2 $\text{Er}_2\text{Ti}_2\text{O}_7$: Weakly Dispersed Crystal Field States

$\text{Er}_2\text{Ti}_2\text{O}_7$ makes an interesting comparison to $\text{Ho}_2\text{Ti}_2\text{O}_7$ as the dispersion of the magnetic excitations is clearly visible to neutron spectroscopy (see Fig. 3.14), indicating magnetic exchange of a few Kelvin [54]. Ignoring this dispersion, the crystal field scheme has been solved by neutron scattering [65]. In its ground state Er^{3+} is roughly the ‘opposite’ of Ho^{3+} having dominant XY anisotropy. The spin wave excitations arising from the exchange coupling can just be resolved by neutron scattering, but a model has not yet been fitted to the data [56].

It is a fairly common practice to ‘extrapolate’ crystal field parameters from one compound to another. However, the comparison of the true crystal field parameters of $\text{Er}_2\text{Ti}_2\text{O}_7$ with those of $\text{Ho}_2\text{Ti}_2\text{O}_7$ shows that this method is quite inaccurate [65]. Thus, considering the parameter B_2^0 , for example, the value for Ho^{3+} is -0.08 and the extrapolated value for Er^{3+} is $+0.08$, but the real value is >0.13 [65]. Whenever possible, neutron scattering should be applied directly to determine the crystal fields.

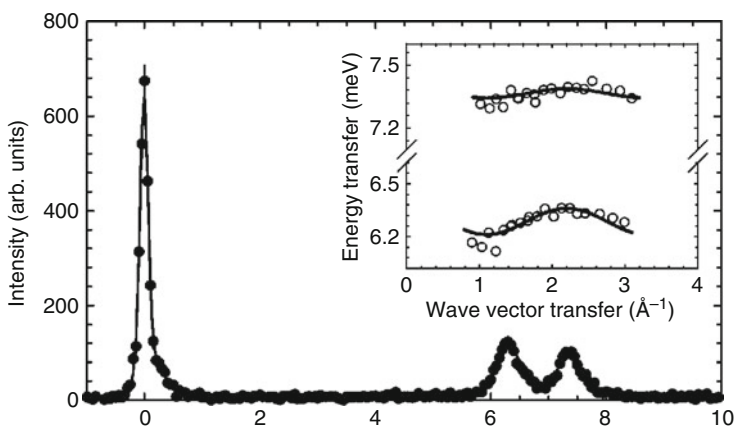


Fig. 3.14 Magnetic excitons in $\text{Er}_2\text{Ti}_2\text{O}_7$ [54]. The main figure shows the powder spectrum and the inset the dispersion of the magnetic excitons at ~ 6 and 7 meV

3.4.3.3 Potassium Iron Jarosite: Spin Wave Dispersions

If a magnet is periodically ordered then neutron scattering affords the most direct way of confirming a spin Hamiltonian and determining its parameters, by measurement of the spin wave dispersion. As regards geometrically frustrated magnets, it is generally necessary that the magnet is ordered, as otherwise the spin waves (if they exist at all) are not classified by the wavevector $\mathbf{Q}$, so their dispersion cannot be resolved. This means that the technique can only be applied to those systems where the frustration is ultimately resolved into magnetic order at low temperature. A good example of this is potassium iron jarosite, for which very detailed spin wave measurements on a single crystal sample [66] and an accompanying theory [67] have been reported.

3.5 Conclusions

In this article, I have tried to illustrate what different magnetic states look like to neutron scattering, and I have emphasised how neutron scattering provides a direct measurement of spin correlation functions, or equivalently, wavevector dependent susceptibilities and magnetic moments. The key figures are Figs. 3.2 and 3.7 which are essentially look-up tables for different magnetic states. A real frustrated magnet might exhibit any of these patterns, either in a pure form or else as a mixture. For example, a frustration-induced suppression of the ordering temperature would generally result in a broad blob of type Fig. 3.2b (broad because the correlations are short) but additional hints of a pattern like Fig. 3.2c would imply that an ice-type rule is operating (but note that not all frustrated magnets obey an ice rule).

A fairly common experience in practical neutron scattering is to prove that some particular material, previously touted as exceptional on the basis of bulk measurements, is in fact highly conventional! The interpretation of bulk measurements inevitably leaves much to the imagination, and neutron scattering, as often as not, provides a sobering correction to those who let their imagination get the better of their scientific caution. For this reason it is always advisable, if possible, to perform neutron scattering on a magnetic substance under investigation. Even if the scattering pattern is poorly resolved, there are usually sufficient statistics to check for consistency with whatever model is being tested. Indeed, neutron scatterers have a saying that ‘neutrons don’t lie’. They mean that the technique is the ideal passive probe of material systems for which the interpretation of experimental results is particularly transparent, robust and, in most cases, unambiguous. Neutron scattering may not be able to image all sorts of magnetic phases, but it remains the most important and direct method of examining the two spin correlation function, so provides the paradigm experiment for the study of highly frustrated magnets.

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